Optimizing Emotion Classification from EEG Signals: A Comparative Analysis of Optimization Techniques on the DEAP Dataset

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Abstract— Electroencephalogram (EEG) signals for emotion classification have gained significant attention in recent times due to its implicit applications in varied fields, including human-computer interaction, affective computing, and substantiated counseling sessions. This study set out to explore the effectiveness of convolutional neural networks (CNNs) in classifying emotions using the DEAP dataset, a publicly available repository of EEG recordings and corresponding emotional annotations by experimenting with different optimizers and loss functions, like "Adamax" and "binary_crossentropy,", etc. By carefully manipulating these optimizers, the study aims to examine and determine the optimal combination with the highest classification accuracy. The findings contribute valuable insights into fine-tuning CNN models for emotion classification based on hyperparameters, paving the way for more robust and effective applications such as personalized diagnosis for counseling, stress-busting techniques and so on. The promising result of this research shows that the "Adamax" optimizer outperforms other optimizers by 82.98 % of the accuracy of classification.

Keywords— Convolutional Neural Network, DEAP dataset, EEG, Emotion Classification, HCI, Hyperparameters, Loss Functions, Machine Learning, Optimizers, Real-time Monitoring

I. Introduction

The domain of emotion analysis and brain-computer interaction using EEG has a vivid scope from a psychological aspect. With advancements in wearable sensors and electrodes, the applications of Human-Brain Interactions are focused by several researchers [1], [2], [3], [4]. The term emotional quotient has gained popularity when it comes to stress handling capacity and ability to deal with one's emotions. The task of emotion classification plays a crucial role in the advances of human resources, and understanding psychology and also in mental disorders[5]. Past research has shown the significant contribution of electroencephalogram EEG signals in emotion classification and state of mind. The EEG device can capture brain waves listed below:

1. Alpha Waves (frequency 8–12 Hz): Calm but alert
2. Beta Waves (frequency 12–30 Hz): Active thinking
3. Theta Waves (frequency 4–8 Hz): Sleep induction, daydreaming, and profound relaxation
4. Delta Waves (frequency 0.5–4 Hz): Profound, dreamless sleep.
5. Gamma waves (frequency above 30 Hz): Higher-order mental functions like perception, problem-solving, and consciousness.
6. Mu waves (frequency 8–13 Hz): When the body is at rest and the mind is quiet.

This research study focuses on potential applications of emotion classification using EEG brain waves signals with a popular classification algorithm, Convolution Neural Network (CNN).

CNN is already being a great choice of researchers for EEG data classification. However, the optimization of CNN is been unexplored and has a challenging research option. This study provides a systematic approach to find an optimal combination of CNN with different optimizers and loss functions.

Section II of this study summarizes a comprehensive literature review in the field of EEG brain wave and classification, followed by section III covering research methods. Section IV is experimental setup whereas section V

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provides a comparative analysis of performance of various optimizers.

II. Literature Review

Deep learning techniques have been a popular choice of EEG classification among past research studies, ranging from emotion analysis, harmony detection to motor imagery and Alzheimer disease detection. [6], [7], [8], [9]. Algorithms Bi-LSTM, with local mean decomposition techniques have been used to achieve 97 % accuracy with specificity of 93.9% in seizure detection[10]. The use of the convolutional neural network (CNN) model using electroencephalogram (EEG) data to recognize emotions is always a popular choice by researchers [11]. The CNN model uses convolution-pooling layer pairs and an output layer, concatenating characteristics from original EEG data into an image format and convolving with one-dimensional filters. The DEAP dataset is used to assess the proposed network similar to the SEED dataset. The DEAP dataset contains 32 participants' EEG data, while SEED includes 15 subjects' EEG data. Similarly, the use of 1-dimensional Convolutional Neural Network (1D-CNN) in combination with other models for emotion categories is explored in another study by combining 1D-CNN-GRU model and 1D-CNN-LSTM models. The 1D-CNN - GRU model finished with validation and check accuracy exceeding 99%. The 1D-CNN-LSTM version outperformed the others with a validation and testing accuracy of 99.9%. The Adam optimizer and the Express-Go entropy loss function were used as hyperparameters. The mode of action demonstrates the high accuracy of the recommended methods in identifying feelings from EEG alerts, with education accuracy exceeding 96% and validation and testing accuracy exceeding 99% [5].

Another study [2] discusses the use of recurrent neural networks (RNN) and gated recurrent units (GRU) in emotion classification based on EEG signals. RNN captures temporal correlations and uses a recurrent process to remember previous inputs. RNN nodes apply a tanh function to keep the values between -1 and 1. GRU, a newer version of recurrent neural networks, uses a bidirectional network to encode the EEG sequence. The layers are the Dense layer, ReLU layer, SoftMax layer, and dropout layer, which are responsible for feature extraction and classification. The ADAM optimizer is used for training the model and updating the parameters after each iteration of the epoch. The proposed model achieves a training accuracy of 96.25% with a learning rate of 0.01 [12]. The research study[13] focused on LSTM hyperparameter tuning where the LSTM-RNN model, a multilabel classification task, was tested using 7500 data and 1000 test data. The model's performance was evaluated using different dropout rates. The model's dropout rate of 0.158 showed more stability in performance. The model's configuration, including hidden neurons, activation functions, and loss functions, was determined based on the problem of recognizing emotion. The LSTM-RNN model was configured with hyperparameters indicated in Fig. 1.

The popular choice of researchers are SEED and DREAMER datasets, [7] have shown high accuracy in emotion recognition using deep learning techniques in research. The Dynamical Graph Convolutional Neural Network (DGCNN) achieved accuracies of 90.4% (subject-dependent) and 79.95% (subject-independent) classification, respectively. The Bag of Deep Features (BoDF) model achieved an accuracy of 93.8% on the SEED dataset and 77.4% on the DEAP dataset. To improve the accuracy, various feature extraction and classification methods were used. Fusion strategies combining multiple modalities achieved accuracies of 71.77% and 73.98%.

NO	Drop out	Accuracy of optimizers (%)			
		Adam	SGD	RMS prop	Ada Grad
1	0.150	0.815	0.336	0.626	0.710
2	0.154	0.671	0.336	0.600	0.693
3	0.158	0.805	0.360	0.609	0.665
4	0.162	0.635	0.348	0.667	0.702
5	0.166	0.682	0.336	0.590	0.714

Fig. 1. The LSTM-RNN model was configured with hyperparameters as discussed in [13].

Combining several optimization techniques in multistage classification for performance improvement is the approach used by researchers. The gradient descent based optimizers are implemented in Deep Neural Networks by [14]. The paper compares the performance of this approach with that of other optimization methods, including SGD, Adam, and Adagrad. It shows that their approach outperforms these methods in terms of accuracy and convergence speed. Additionally, the authors conduct sensitivity analysis to evaluate the robustness of their approach to implementations of multistage optimization techniques that perform much better. Such that using SGD gives 6.8706 loss and 0.4875 Accuracy whereas the combination of SGD and Adam gives 6.1812 loss and 0.7310 Accuracy [1].

Feature extraction plays an important role in EEG classification. In this paper by Nakra et al., the authors propose a methodology for classifying motor imagery (MI) tasks using statistical wavelet features based on sub-band selection. The approach utilizes the harmony search optimization algorithm for feature selection and a Long Short-Term Memory (LSTM) neural network for classification. The study employs various techniques such as frequency domain analysis, band power evaluation, autoregressive parameter estimation, and sensitivity index evaluation. The results of the study demonstrate a significant enhancement in classification accuracy, with the reported standard deviation value of the EEG signal sub-band being r = 1.23. The following shows that the proposed method achieves an accuracy of 97.92% with a sensitivity of 100% and specificity of 96.43%.[2]. Another research for classifying motor imagery EEG signals using a combination of deep short-term memory (LSTM) networks and neighbour component analysis (NCA). Using LSTM DNN to preprocess and extract EEG signal features, along with considering mean square error (MSE) and cross-entropy criterion, helps achieve over 95% accuracy in EEG classification. The article emphasizes the importance of using a

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wrapper approach for feature selection to reduce processing time and avoid overfitting of deep neural networks (DNN). Taking into account alpha and beta waves in the analysis further strengthens the usefulness of the proposed method. The obtained accuracy above 95 demonstrates the effectiveness of the proposed method in classifying motor imagery EEG signals. The combination of LSTM and NCA in the proposed method shows promising results in EEG signal classification, highlighting its potential for practical applications of brain-computer interfaces. The accuracy obtained by the proposed method demonstrates its effectiveness in the task of EEG classification [5]. SVM and KNN classifiers for classification consists of 150 data samples from 100 patients, with 2/3rd data samples used for training and 1/3rd data samples remaining for testing. The aim of the research is to provide better results in terms of classification rate and diagnosis accuracy. classification rate as well as diagnostic accuracy obtained is more and it provides comparatively better results than individuals in terms of accuracy (up to 94%) [3]. The article [4] presents a new approach to classifying epileptic seizures using deep learning techniques. The proposed method is based on a bidirectional long short-term memory (Bi-LSTM) network, which integrates information before and after the current scan time to decide the output state.

The proposed study focuses on the different types of optimizers, examining their impact on the efficiency and adaptability of convolutional neural networks (CNNs) using the DEAP dataset. These optimizers demonstrate crucial attributes, such as adaptive learning rates and moment estimates, contributing to rapid convergence during training—a critical factor for real-time processing. The optimizer's adaptability to dynamic gradients enhances the model's ability to discern subtle changes in emotional states in real-time, ensuring accurate and timely emotion analysis. Beyond training efficiency, optimizers influence the model's generalization capability and energy efficiency, making their careful selection imperative for deploying reliable and resource-conscious emotion analysis systems in real-world applications.

III. Research Methods

This research methodology includes four processes: data preprocessing, feature extraction, feature matrix generation, and the use of a CNN classifier. The details of each step are further explained (Fig. 2.).

A. Data Collection (DEAP Dataset)

The dataset used in this study is DEAP, i.e. it is used by researchers to test their effective state estimation method. This dataset contains EEG and other peripheral physiological signals [15]. The proposed research study aims to focus on the effectiveness of Convolution Neural Network (CNN) models in emotion classification using a publicly available dataset - DEAP. We propose an iterative approach of using a total of 10,240 records of a random subset of 8 participants from the DEAP dataset that involves training and evaluating CNN models with different optimizers and loss functions. The optimizer guides the learning process of weights updation of the CNN based on the error between the predicted and actual outputs. Introduction of the following 7 optimizers: "Adam", “RMSprop”, “Adagrad”, “Adadelta”, “Adamax”, “Nadam” and "SGD" cause the results to fluctuate. The loss function guides the learning process by measuring the discrepancy between the predicted and actual outputs. We performed experimentation in running all 7 different types of optimizers with 2 different types of loss functions: “binary_crossentropy” and “categorical_crossentropy”.

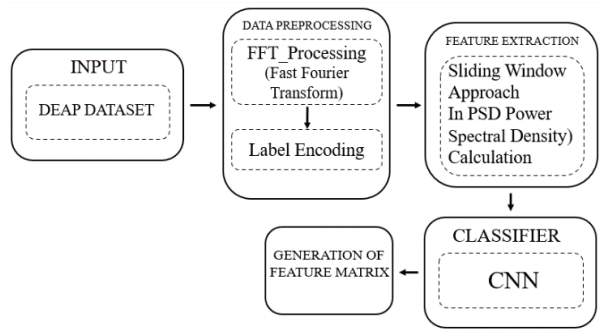

Fig. 2. Block Diagram of our proposed method

A. Data Preprocessing with Fast Fourier Transform and Label Encoding

A crucial signal processing method for mining electroencephalogram (EEG) continuous signal information is Fast Fourier Transform (FFT) [16]. Mathematically, the FFT is expressed as

$$f(x)=a_0+\sum_{n=1}^{\infty}\left(a_n\cos\frac{n\pi x}{L}+b_n\sin\frac{n\pi x}{L}\right) \quad (1)$$

FFT features are followed by label encoding

$$f: \text{Categories} : \text{Integers} \quad (2)$$

B. Feature Extraction

The proposed research study uses the Sliding Window Approach in Power Spectral Density calculation. Periodogram and power spectral viscosity computations and combining band waves of varied frequencies are needed for point extraction with the help of FFT.

The formulae for the same are as follows:

$$P(f)=\frac{1}{MU}\left|\sum_{L=0}^{U-1} x_i\,(n)e^{2\Pi f}\right|^2 \quad (3)$$

C. Generation of Feature Matrix

The construction of a robust feature matrix is a foundational step in our methodology, aiming to distill salient information from the pre-processed DEAP dataset for subsequent input into the convolutional neural network (CNN) models. Each EEG recording, post-pre-processing, undergoes feature extraction to encapsulate key characteristics which are crucial for emotion classification. Specifically, spectral power

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features are computed for distinct frequency bands, utilizing the Fast Fourier Transform (FFT)

Additionally, statistical features, as follows, are calculated to capture temporal characteristics:

$$Mean = \frac{1}{N}\sum_{n=0}^{N-1} x_n \qquad (4)$$

$$Std\ Dev = \sqrt{\frac{1}{N}\sum_{n=0}^{N-1}(x_n - mean)^2} \qquad (5)$$

$$\text{Skewness} = \frac{\sum_{n=0}^{N-1}(x_n - mean)^3}{(N-1)\ Std\ Dev^3} \qquad (6)$$

$$\text{Kurtosis} = \frac{\sum_{n=0}^{N-1}(x_n - mean)^4}{\sum_{n=0}^{N-1}(x_n - mean^2)^2} \qquad (7)$$

The amalgamation of these spectral and statistical features results in a comprehensive feature matrix, where each row corresponds to an EEG recording, and each column represents a distinct feature. This matrix, denoted as X, is instrumental in training and evaluating the CNN models for emotion classification. The feature matrix is then normalized to mitigate variations and enhance model convergence, ensuring a robust foundation for subsequent analyses and model optimizations.

D. CNN Classifier

The architecture of CNN often integrates multiple convolutional and pooling layers to create a hierarchy of features, allowing the network to discern intricate patterns. Following these layers, densely connected layers aggregate the learned features and perform the final classification.

Mathematically, the convolutional operation can be expressed as,

$$G = H * F\ that\ is$$

$$G[i,j] = \sum_{u=-k}^{k}\sum_{u=-k}^{k} H[u,v].F[i-u\ ,j-v] \qquad (8)$$

The pooling operation is typically realized through max-pooling, where the maximum value within a defined window is retained. This operation is expressed as,

$$P(i,j) = [\![max]\!]\ (m,n)\ I(i\cdot s+m, j\cdot s+n) \qquad (9)$$

IV. Experimental Setup

For this research, we have used 10,240 records from the preprocessed version of the DEAP dataset. (Table I). A randomized stratification approach was employed to partition the dataset into distinct sets for training and testing, maintaining a balanced distribution, and adhering to a ratio of 75:25, respectively. The resultant data configuration for our experimentation is elucidated in the subsequent section, encapsulating the structural dimensions essential for the rigorous evaluation and optimization of the proposed convolutional neural network (CNN) model.

TABLE I. Dataset Dimensions

Set	Data	Target	%
Train Set	(124928, 70)	(124928, 4)	75%
Test Set	(32132, 70)	(31232, 4)	25%

E. Hyperparameters

In order to achieve optimal model performance, proper tuning of hyperparameters is required for which the chosen batch size is set to 256. Several optimizers are employed to facilitate efficient weight updates during backpropagation. Given the multi-labelled nature of the dataset, the binary cross-entropy and categorical cross-entropy as loss functions are implemented to guide the model training process (Table II).

TABLE II. Hyperparameters

Parameter	Value
Batch size	256
Optimizer	Adam, Adamax, Nadam, SGD, RMSprop, Adagrad, Adadelta
Loss Functions t	Categorical cross-entropy, binary cross-entropy

Table III shows the different types of optimizers whereas Table IV shows loss functions used in this study:

TABLE III. List of Optimizers

Optimizer	Mathematical Formulae
Adam	$\theta_{t+1} = \theta_t - \frac{\tau}{\sqrt{\hat{\vartheta}_t + \epsilon}} * \widehat{m_t}$
Adamax	$\theta t+1 = \theta t - \frac{\alpha}{max(\frac{\beta^{t_1}}{\beta 2})} \odot mt$
Nadam	$\theta t+1 = \theta t - \frac{\alpha}{\sqrt{\hat{v}}+\epsilon}(\beta 1mt + (1-\beta 1) \nabla L(\theta t))$
SGD	$\theta t + 1 = \theta t - \alpha \nabla L(\theta t)$
RMSprop	$\theta t + 1 = \theta t - \frac{\alpha}{\sqrt{E[g^2]t + \epsilon}} \nabla L(\theta t)$
Adgrad	$\theta t + 1 = \theta t - \frac{\alpha}{\sqrt{Gt + \epsilon}} \nabla L(\theta t)$
Adadelta	$\theta t+1 = \theta t - \frac{\sqrt{\Delta xt + \epsilon}}{\sqrt{Gt+\epsilon}} \nabla L(\theta t)$

The following Table IV displays the types of loss functions that have been incorporated into our research:

TABLE IV. Loss Functions

Loss Function	Mathematical Formulae
Binary Cross-Entropy	$-\frac{1}{N}\sum_{i=1}^{N}[yi\ log(pi) + (1-yi)log(1-pi)]$
Categorical Cross-Entropy	$-\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} yi,jlog(pi,j)$

V. Implementation and Results

This experiment aims to find the optimal combination of optimizers with the corresponding loss functions to achieve the

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highest performance of emotion classification in our proposed model. The optimization techniques experimented with are Optimizer Adaptive moment estimation ((Adam), Adaptive moment estimation (Adamax), Nesterov Adam (Nadam), Stochastic gradient descent (SGD), quadratic propagation (RMSprop), Adaptive gradient algorithm (Adagrad), and lastly Adaptive delta, (Adadelta). The performance of each optimizer was determined by varying the loss functions, i.e. H. the binary cross entropy and the categorical cross entropy, experimented. The experiment proposed in this study used 15 layers with 200 epochs and 256 batch sizes. The following Table V show the result of this experiment.

TABLE V. RESULTS

Optimizer	**Accuracy with Loss Functions (%)**	
	Binary Cross-Entropy	***Categorical Cross-Entropy***
Adam	0.8143	0.7808
Adamax	0.8210	0/7993
Nadam	0.8160	0.7858
SGD	0.5091	0.7707
RMSprop	0.7918	0.7557
Adagrad	0.3677	0.5183
Adadelta	0.3391	0.3916

VI. DISCUSSION

In this section, we are presenting the results of our experiment. As mentioned above and shown in the table, we executed combinations of different optimizers with 2 different types of loss functions. As displayed in the table, the top 3 optimizers are Adam, Adamax and Nadam and the loss function that displays the maximum accuracy is Binary Cross-entropy. Although with categorical cross-entropy the top 3 optimizers with the best accuracy were the same but they obtained lesser accuracy as compared to binary cross-entropy.

T									
0	1700	25	65	25	32	39	22	22	220
1	110	1100	81	16	42	38	78	11	1
2	23	16	1700	61	52	96	53	34	10
3	28	14	83	1300	130	98	77	18	9
4	78	23	72	89	3200	180	130	37	56
5	75	14	110	130	200	5300	260	55	130
6	200	71	170	210	270	430	6800	200	360
7	49	18	69	47	86	93	140	2800	79
8	330	1	70	16	13	32	36	60	1000
°P	0	1	2	3	4	5	6	7	8

Fig. 3. Confusion Matrix of Adamax optimizer with binary cross-entropy

With binary cross-entropy, Adam gives an accuracy of 81.43%, Adamax gives an accuracy of 82.10% and Nadam gives an accuracy of 81.60%. These optimizers and binary cross-entropy show the best performance.

The following figures (Fig. 3-.6) show the required accuracy change per epoch. The algorithm does not possess any overfitting and the training process stops at 200 epochs. 5 to 10 seconds average training time for each epoch is observed. The confusion matrix to analyze the difference between predicted and actual values is also shown in below figures.

T									
0	1700	25	65	25	32	39	22	22	220
1	110	1100	81	16	42	38	78	11	1
2	23	16	1700	61	52	96	53	34	10
3	28	14	83	1300	130	98	77	18	9
4	78	23	72	89	3200	180	130	37	56
5	75	14	110	130	200	5300	260	55	130
6	200	71	170	210	270	430	6800	200	360
7	49	18	69	47	86	93	140	2800	79
8	330	1	70	16	13	32	36	60	1000
°P	0	1	2	3	4	5	6	7	8

Fig. 4. Confusion Matrix of Nadam optimizer with binary cross-entropy

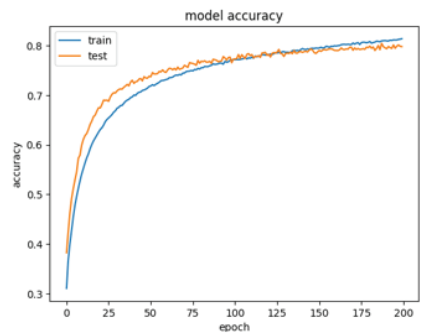

Fig. 5. Model Accuracy of Adam optimizer with binary cross-entropy

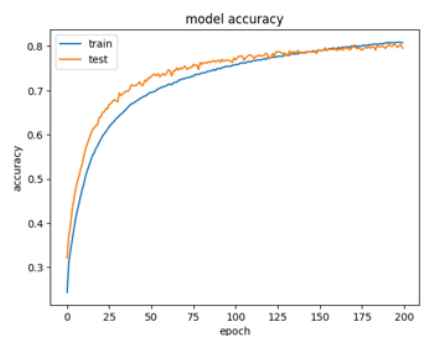

Fig. 6. Model Accuracy of Nadam optimizer with binary cross-entropy

VII. CONCLUSION

In this paper, we developed a CNN model for the classification of EEG emotion. We conducted an experiment to obtain the optimal combination of hyperparameters for the model. The best generalization of 0.8210% accuracy was obtained with Adamax and binary cross-entropy. This result was compared against 6 other optimizers with categorical

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cross-entropy and binary cross-entropy and observed the best accuracy at 0.8210% with Adamax with binary cross-entropy while Adagrad with binary cross-entropy performed the worst at 0.3677% accuracy.

The successful deployment of the Convolutional Neural Network (CNN) architecture for emotion classification based on electroencephalogram (EEG) data paves the way for exciting future research endeavors and practical applications. Notably, the real-time implementation on hardware platforms, and subsequent steps involving compiling a dedicated database derived from acquired EEG signals and executing comprehensive evaluations utilizing the established model, hold immense potential for ground-breaking advancements in this domain.

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